clk

reset

ex_mem_reg

ex_mem_pipeline

clk enable mem_read_in mem_read_out mem_write_in mem_write_out
reg_write_en_in

reg_write_en_out

reset

alu_result_in[31:0] alu_result_out[31:0]

pc_4_in[31:0] pc_4_out[31:0]
rd_addr_in[4:0] rd_addr_out[4:0]

rs2_data_in[31:0]

rs2_data_out[31:0] wb_sel_in[1:0] wb_sel_out[1:0]

dmem_inst

data_memory

clk mem_read mem_write

alu_result[31:0] dmem_data[31:0] write_data[31:0] mem_wb_reg

mem_wb_pipeline

clk enable

reg_write_en_in

reg_write_en_out

reset

alu_result_in[31:0] alu_result_out[31:0] dmem_data_in[31:0] dmem_data_out[31:0] pc_4_in[31:0] pc_4_out[31:0]

rd_addr_in[4:0] rd_addr_out[4:0]

wb_sel_in[1:0] wb_sel_out[1:0]

wb_mux_inst

writeback_mux

alu_result[31:0] dmem_data[31:0] pc_4[31:0] wb_sel[1:0] write_data[31:0]

fwd_inst

forwarding_unit

ex_mem_reg_write_en

mem_wb_reg_write_en

ex_mem_rd_addr[4:0]

forwardA[1:0]
forwardB[1:0]

id_ex_rs1_addr[4:0]

id_ex_rs2_addr[4:0] mem_wb_rd_addr[4:0]

op_mux_inst

operand_selection_muxes

op1_sel op2_sel

ex_mem_alu_result[31:0]
forwardA[1:0]
forwardB[1:0]

imm[31:0] mem_wb_write_data[31:0] operand_a[31:0] operand_b[31:0] pc[31:0] rs1_data[31:0] rs2_data[31:0]

branch_cmp_inst

branch_comparator

branch_taken

branch_jump[2:0] rs1_data[31:0] rs2_data[31:0]

pc_control_inst

pc_control branch_taken

branch_jump[2:0] pc_sel[1:0]

branch_adder_inst

branch_target_adder

imm[31:0] pc[31:0]

target_address[31:0]

pc_4_adder_inst pc_4_adder pc[31:0] pc_4[31:0]

pc_mux_inst

pc_mux

alu_result[31:0] next_pc[31:0] pc_4[31:0] pc_sel[1:0]

target_address[31:0]

!=

branch_flush0_i RTL_NEQ O

I0[2:0]
I1[2:0]

pc_reg

pc

clk enable

reset next_pc[31:0] pc[31:0]

branch_flush_i RTL_AND

I0

I1 O

imem_inst

instruction_memory

pc[31:0] instruction[31:0]

if_id_reg

if_id_pipeline

clk enable
flush

reset

instruction_in[31:0]

instruction_out[31:0]

pc_4_in[31:0] pc_in[31:0] pc_4_out[31:0] pc_out[31:0]

ctrl_inst

main_control_unit

mem_read mem_write

op1_sel op2_sel reg_write_en

alu_op[4:0] branch_jump[2:0]

funct3[2:0]
funct7[6:0]

imm_sel[2:0]

opcode[6:0] wb_sel[1:0]

hdu_inst

hazard_detection_unit

id_ex_mem_read id_ex_flush
if_id_enable
pc_enable

id_ex_rd_addr[4:0]
id_rs1_addr[4:0]
id_rs2_addr[4:0]

imm_gen_inst

immediate_generator

imm_sel[2:0] imm[31:0]
instruction[31:0]

rf_inst

register_file

clk

reg_write_en

reset

rd_addr[4:0]

rs1_addr[4:0]

rs1_data[31:0]

rs2_addr[4:0]

rs2_data[31:0] write_data[31:0]

id_ex_flush_i RTL_OR

I0

I1 O

id_ex_reg

id_ex_pipeline

clk enable

flush mem_read_in mem_read_out mem_write_in mem_write_out op1_sel_in

op1_sel_out op2_sel_in

op2_sel_out

reg_write_en_in

reg_write_en_out

reset

alu_op_in[4:0] alu_op_out[4:0] branch_jump_in[2:0] branch_jump_out[2:0]
imm_in[31:0]

imm_out[31:0]

pc_4_in[31:0] pc_in[31:0] pc_4_out[31:0] pc_out[31:0]

rd_addr_in[4:0] rd_addr_out[4:0]

rs1_addr_in[4:0]

rs1_addr_out[4:0]

rs1_data_in[31:0]

rs1_data_out[31:0]

rs2_addr_in[4:0]

rs2_addr_out[4:0]

rs2_data_in[31:0]

rs2_data_out[31:0] wb_sel_in[1:0] wb_sel_out[1:0]

alu_inst

alu

alu_op[4:0] operand_a[31:0] alu_result[31:0] operand_b[31:0]

led[15:0]